

TAJAMUL ASHRAF

Homepage [◇](#) [Google Scholar](#) [◇](#) [GitHub](#) [◇](#) [LinkedIn](#) [◇](#) [☎](#) +1(443)-682-5437 [◇](#) [✉](#) tashraf@jhu.edu

RESEARCH INTERESTS

Multimodal Agents, Vision-Language Reasoning, Medical AI, Video Understanding, Embodied AI, Visual Grounding, Domain Generalization, and AI for Science

EDUCATION

Johns Hopkins University

PhD in Computer Science

Advisor: **Prof. Vishal Patel** and **Prof. Daniel Khashabi**

Thesis: *Agentic Multimodal Reasoning for Cataract Surgery Videos*

Sep 2026 – Present

Baltimore, USA

Indian Institute of Technology Delhi

Master's (Research) in Information Technology, GPA: 9.28/10.0

Advisor: **Prof. Chetan Arora**

Thesis: *Domain Adaptation in Breast Cancer Detection from Mammograms.*

Aug 2022 – June 2024

New Delhi, India

National Institute of Technology Srinagar

B.Tech. in Information Technology, GPA: 8.34/10.0

Advisor: **Prof. Janibul Bashir**

Thesis: *Robust DNNs for Analyzing the Impact of Coherent Climatic Factors.*

Aug 2018 – July 2022

Srinagar, India

SELECTED PUBLICATIONS

 Best Paper Award, * equal contribution, † my role as mentor

- J2** Komal Kumar*, **Tajamul Ashraf***, Omkar Thawakar, Rao Muhammad Anwer, Hisham Cholakkal, Mubarak Shah, Ming-Hsuan Yang, Phillip Torr, Salman Khan, Fahad Khan, “*LLM Post-Training: A Deep Dive into Reasoning Large Language Models.*” In **IEEE TPAMI**. [📄 Paper](#), [🔗 Code](#)
- C9** Rozain Shakeel, Abdul Rahman, Muneeb Mushtaq, Tausifa Jan Saleem, **Tajamul Ashraf†**, “*MedSPOT: A Workflow-Aware Sequential Grounding Benchmark for Clinical GUI.*” In **ECCV’26**. [🔗 Code](#) [📄 Paper](#)
- J1** **Tajamul Ashraf**, Rajes Manna, Partha Purkayastha, Tawaheed Tariq, Janibul Bashir, “*Context Aware Grounded Teacher for Source Free Object Detection.*” In **IJCV**. [🔗 Code](#) [📄 Paper](#)
- C8** Akash Ghosh, **Tajamul Ashraf†**, Rishu Singh, Numaan Saeed, Sriparna Saha, Xiuying Chen, Salman Khan, “*CarePilot: A Multi-Agent Framework for Long-Horizon Computer Task Automation in Health-care.*” In **CVPR’26 Findings**. [🔗 Code](#) [📄 Paper](#)
- C7** **Tajamul Ashraf***, Amal Saqib*, Jean Lahoud, Hisham Cholakkal, Mubarak Shah, Philip Torr, Fahad Khan, Rao Anwer, Salman Khan, “*Agent-X: Evaluating Deep Multimodal Reasoning in Vision-Centric Agentic Tasks.*” In **ICLR’26**. [🔗 Code](#) [📄 Paper](#) [📄 Data](#)
- C6** **Tajamul Ashraf**, Janibul Bashir, “*TITAN: Query-Token based Domain Adaptive Adversarial Learning.*” In **ICCV 2025**. [🔗 Code](#) [📄 Paper](#)
- C5** Jinhong Wang*, **Tajamul Ashraf*†**, Zongyan Han, Jorma Laaksonen, Rao M. Anwer, “*MIRA: A Novel Framework for Fusing Modalities in Medical RAG.*” In **ACM Multimedia 2025**. [🔗 Code](#) [📄 Paper](#)
- C4** Bhuvan Sachdeva, Naren Akash, **Tajamul Ashraf**, Simon Mueller, Thomas Schultz, Maximilian W. M. Wintergerst, Niharika S. Prasad, Kaushik Murali, Mohit Jain, “*Phase-Informed Tool Segmentation for Manual Small-Incision Cataract Surgery.*” In **MICCAI 2025**. [📄 Paper](#)
- C3** **Tajamul Ashraf**, Krithika Rangarajan, Mohit Gambhir, Richa Gauba, Chetan Arora, “*D-MASTER: Mask Annealed Transformer for Unsupervised Domain Adaptation in Breast Cancer Detection from Mammograms.*” In **MICCAI 2024**. [🔗 Code](#) [📄 Paper](#)

- C2** 🏆 **Tajamul Ashraf**, Tisha Madame, “*HF-Fed: Hierarchical Based Customized Federated Learning Framework for X-Ray Imaging.*” In **MICCAI 2024** Workshop (Deep-Brea³th). [🔗 Code](#) [📄 Paper](#)
- C1** **Tajamul Ashraf**, Fuzayil Mir, Iqra Altaf Gillani, “*TransFed: A Way to Epitomize Focal Modulation Using Transformer-Based Federated Learning.*” In **WACV 2024**. [🔗 Code](#) [📄 Paper](#)

RESEARCH EXPERIENCE

Johns Hopkins University Sep 2026 – Present
Research Assistant Hosts: Prof. Daniel Khashabi and Prof. Vishal Patel Baltimore, USA

- Developing video-to-language agents that reason over temporal surgical events, track procedural state, and produce grounded, verifiable feedback.

Nanyang Technological University (NTU) Mar 2026 – Aug 2026
Research Manager Host: Prof. Adams Wai Kin Kong Nanyang, Singapore

- Collaborated with the **Interpol** team on video understanding and reasoning for surveillance.
- Developed multimodal agentic systems for long-horizon temporal reasoning and decision-making.

King Abdullah University of Science and Technology (KAUST) Dec 2025 – Mar 2026
Research Consultant Host: Prof. Bernard Ghanem Remote

- Introduced **MedCTA**, a benchmark for evaluating multimodal clinical tool agents.
- Developed evaluation protocols for grounded and reliable tool-augmented clinical reasoning.

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI) Jan 2025 – Nov 2025
Research Engineer Hosts: Prof. Salman Khan, Prof. Rao Anwer Abu Dhabi, UAE

- Curated **Agent-X**, a large-scale benchmark for perception, grounding, and decision-making in real-world agentic tasks; Awarded **2nd Place** (Research Track) at the Agentic AI Summit, UC Berkeley (2025).
- Built a multimodal RAG framework (**MIRA**) for factual and adaptive medical agents.
- Authored a survey on post-training techniques for reasoning LLMs.

Microsoft Research (MSR) July 2024 – Dec 2024
Research Intern Hosts: Dr. Mohit Jain, Dr. Prachi Jain Bangalore, India

- Built LLM-based Knowledge-Graph with RAG framework on a semi-structured KB for medical chatbots.
- Jointly introduced *Cataract-MSICS* dataset and proposed a phase-conditioned method with pseudo-labeling for surgical tool segmentation.

AI & Robotics Technology Park (ARTPARK), I-Hub Dec 2021 – May 2022
Visiting Student Host: Prof. Raghu Krishnapuram Bangalore, India

- Developed an automated washroom-cleaning robot *RoboJanitor*; implemented pose estimation, detection, classification, and SLAM.
- Built 3D pose models and REST APIs for production deployment.

Indian Institute of Science (IISc) Mar 2021 – Sep 2021
Research Intern Host: Prof. S. N. Omkar Bangalore, India

- Built a human-pose comparison model to assist yoga instruction in live online settings.
- Used BlazePose for keypoint extraction and delivered a feedback mechanism for real-time pose correction.

ACADEMIC SERVICES

Teaching Assistant

Probability and Statistics (SC2000), NTU Singapore (Spring 2026), instructed by **Prof. Adams Kong**.

Trustworthy Medical Vision (CV8502), MBZUAI (Fall 2025), instructed by **Prof. Yutong Xie**.

Computer Vision for Robotics (JRL780), IIT Delhi (Spring 2024), instructed by **Prof. Chetan Arora**.

Computer Science Fundamentals and Programming (COL100), IIT Delhi (Fall 2023), instructed by **Prof. Chetan Arora** and **Prof. Vireshwar Kumar**.

Data Structures and Algorithms (COL106), IIT Delhi (Fall 2022), instructed by **Prof. Naveen Garg** and **Prof. Ashish Chiplunkar**.

Reviewer

2027	ICLR, IJCV
2026	NeurIPS, ACM Multimedia, MICCAI, IJCV
2025	ICCV, ACM Multimedia, MICCAI, WACV, Pattern Recognition, IJCV, TAI
2024	MICCAI, WACV, CVIP, TPDS
2023	CVIP, HPEC

Volunteering

Program Committee Member: ACM Multimedia 2026 (Rio de Janeiro, Brazil).

Volunteer: ICLR 2026 (Rio de Janeiro, Brazil), ACM COMPASS 2025 (Toronto, Canada), CVIP 2023 (IIT Jammu, India).

Invited Talks

Invited speaker at *Sangam 2.0* (NIT Srinagar Alumni Meet), delivering a talk on *Research Directions in current AI landscape*, covering research opportunities, higher studies, and emerging trends in AI (2025).

Delivered a keynote talk on “Multimodal Systems for Vision-Language Domains in Advancing Healthcare” at the BRAIN-AID Seminar, University of Lagos, Nigeria (April 2025)

Gave two talks on undergrad research directions and campus placements at NIT Srinagar (June 2024)

Mentorship

To give back, I co-founded Gaash Research Lab to foster research culture among students and actively mentor Bachelor’s and Master’s students on thesis projects and publication-oriented research. Non-exhaustive list:

2026

14. Ilakkiya Selvakumar (NTU Singapore) *Benchmarking Multimodal Agents for Surveillance Applications*
13. Uzair Nazir, Abdul Muizz (NIT Srinagar) *OCR and kashmir language translation*
12. Adab Ismail (NIT Srinagar) *Vision language modeling for object tracking*
11. Sheikh Samee ul Haq, Aabid Nabi (NIT Srinagar) *Robust Tool Segmentation in Surgical Scenarios*
10. Rozain Shakeel, Muneeb Ahmad (NIT Srinagar) *Grounding and Reasoning for Clinical GUI*
9. Tafheem Malik, Iram Naseer (NIT Srinagar) *Multimodal Image and Video Generation using Reasoning*
8. Ifrah Mushtaq (University of Kashmir) *ASR and Phonetic Analysis for Kashmiri Language*

2025

7. Burhaan Rasheed Zargar (NIT Srinagar) *Script-Aware Flow Matching for Kashmiri Text-to-Speech*
5. Abrar Riyaz (NIT Srinagar), Wasif Tak (Thapar University) *Language-Conditioned Surgical Tool Segmentation*
4. Tawaheed Tariq, Sonia Yadav (NIT Srinagar) *Query-Driven Reasoning for Multi-modal MOT*

2024

3. Suhaib Salmani (NIT Srinagar) *VLM-based Masked Learning for Efficient Medical Object Detection*
2. Rajes Manna (NIT Srinagar) *Context Aware Grounded Teacher for Source-Free Object Detection*

2023

1. Asrar ul Haq (NIT Srinagar) *Transformer-Based Unsupervised Domain Adaptation in a Federated Setup*

ACHIEVEMENTS AND HONORS

- Awarded the **Johns Hopkins PhD Fellowship** and **Summer Relocation Grant**, (Baltimore, USA).
- **ICLR 2026 Registration and Travel Grant** (Rio de Janeiro, Brazil).
- **MICCAI 2025 Outstanding Reviewer Award**.
- Awarded travel and registration grants for **ICCV 2025** (Honolulu, Hawaii), **ACM Multimedia 2025** (Dublin, Ireland), and the **Agentic AI Summit 2025** (UC Berkeley, California).
- **Second Place** in the Agent-X Competition at the Agentic AI Summit 2025, UC Berkeley.
- Secured 2nd place in the MBZUAI Department Logo Design Competition.
- Awarded a \$500 compute grant from **alphaXiv** for contributions to LLM research.
- **Best Student Paper Award**, Deep-Brea³th Workshop, MICCAI 2024 (HF-Fed).
- Nominated as a **Diversity Intern** by Microsoft Canada.
- Awarded the **IIT Delhi Endowment Grant** and travel grant for **MICCAI 2024** (Marrakesh, Morocco).
- Gold **Microsoft Student Learn Ambassador** at NIT Srinagar.
- Won India's largest **National Entrepreneurship Challenge** at IIT Bombay (100,000 INR cash prize).

MEDIA APPEARANCES

- *NIT Srinagar Hosts KATHE 2026 Challenge to Develop English-to-Kashmiri Translation.* [🔗](#)
- *A New Stress Test for AI Agents That Plan, Look, and Click.* [🔗](#)
- *Meet a Microsoft Learn Student Ambassador Graduate: Tajamul Ashraf.* [🔗](#)
- *NIT Srinagar Students Win India's Largest National Entrepreneurship Challenge.* [🔗](#)
- *Hult Prize: A Futuristic Approach to Entrepreneurship.* [🔗](#)
- *Workshop on Entrepreneurship, Research and Innovation at NIT Srinagar.* [🔗](#)

SKILLS & INTERESTS

Technical Skills	Python, C/C++, PyTorch, DeepSpeed, FSDP, Model/Sequence Parallelism
Leadership / Outreach	Founder of Ralith Milth (anti-drug NGO), Co-founder Gaash Lab (Mentoring)
Personal Interests	Badminton, swimming, blog (Journey with Bytes)

REFERENCES

[Prof. Adams Kong](#) (NTU Singapore) [Prof. Daniel Khashabi](#) (Johns Hopkins) [Prof. Vishal Patel](#) (Johns Hopkins)